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Neural network for predicting satellite failures using the ESA dataset

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1. Introduction and Motivation

The aerospace domain harsh environment conditions impose several challenges to satellite operations. In order to ensure mission success and longevity, failure prediction and detection mechanisms show an essential role. In the literature, neural networks, particularly neuron networks, have demonstrated promising results in fault-related metrics, such as Time-to-Failure (TTF) prediction and Early Fault Detection (EFD) (Fang; Nagarajan, 2004).

However, few existing works emphasize the constraints of running these models on embedded devices, either via software (AI accelerators) or hardware (FPGAs). These devices need to be radiation-hardened, low power, and, consequently, with limited computational resources, while at the same time need near real-time results (low latency) (Diana; Dini, 2024). These constraints alone pose significant challenges for deploying robust machine learning models On-Board Computers (OBCs).

Another known challenge in the field is the lack of publicly available datasets for satellite telemetry data with labeled failures. Luckily, the recent release of the ESA-ADB dataset (De Canio; Kotowski; Haskamp, 2024) opens new opportunities for research in this area. The dataset contains multivariate time series data from various satellite missions, providing a benchmark for anomaly detection tasks. Due to its youth, the dataset is still rarely explored in the literature, which further motivates this project.

With that in mind, the goal of this master's program is to develop a neural network for predicting satellite failures, initially exploring the ESA dataset (ESA Anomaly Dataset, *[n.d.]*) and subsequently deploying the model on edge devices compatible with satellite OBCs. The primary anticipated challenges include:

- Nature and quality of the dataset: The dataset contains real-world satellite telemetry data, which may include noise, missing values, and imbalanced classes due to the rarity of failures.
- Model generalization: Each satellite mission may have different characteristics, making it challenging to develop a model that generalizes well across various scenarios.
- Hardware constraints: The model needs to be optimized for deployment on resource-constrained edge devices, requiring careful consideration of model size, inference time, and power consumption.
- Model interpretability: In safety-critical applications like satellite operations, it is essential to understand the model's decision-making process to ensure trustworthiness and reliability.

2. State of the Art

This section provides an overview of the current state of the art in satellite telemetry data analysis and fault detection.

Initially released in mid-2024, with an updated version in 2025, the European Space Agency (ESA) Anomaly Dataset (De Canio; Kotowski; Haskamp, 2024) poses as a fresh, large-scale, real-life satellite telemetry dataset with annotated anomalies from three ESA missions. Out of these three missions, two are included in the ESA-ADB (Anomaly Detection Benchmark) project (Kotowski *et al.*, 2024), which provides a baseline for evaluating anomaly detection algorithms on satellite telemetry data. The work describes the dataset in detail and presents some unsupervised and semi-supervised time series anomaly detection (TSAD) algorithms using the TimeEval framework (Wenig; Schmidl; Papenbrock, 2022).

The novelty of the ESA-ADB dataset means that few works have explored it so far. (Cena *et al.*, 2026) proposes a self-supervised learning approach for anomaly detection in satellite telemetry data. The model is based on Real-valued Non-Volume Preserving (Real NVP) architecture, tailored for performance but also optimized for deployment on edge devices. The model's generalization capabilities are evaluated by testing it on multiple datasets: the ESA-ADB (Kotowski *et al.*, 2024), ADAPT (ADAPT Dataset - NASA Open Data Portal, *[n.d.]*), and SMAP and MSL (Hundman *et al.*, 2018).

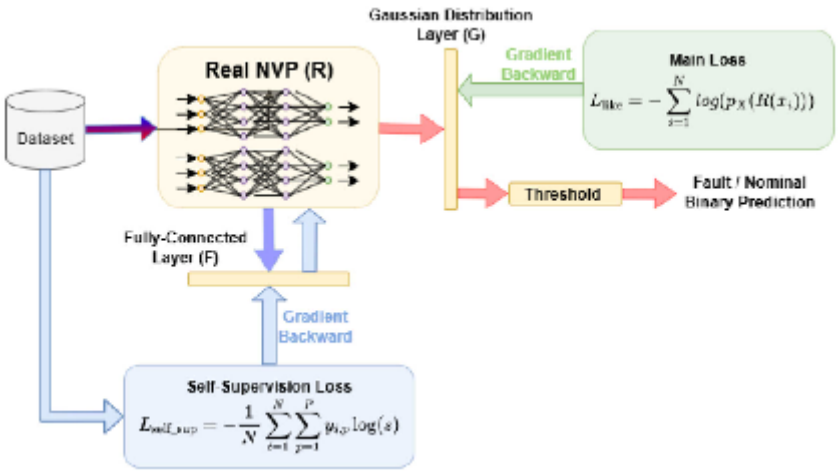

Figure 1: Process pipeline proposed by (Cena *et al.*, 2026).

(Kiprit *et al.*, 2024) explores the use of Graph Neural Networks (GNNs) for anomaly detection in spacecraft, also highlighting the goal of deploying the model on space-grade hardware. The work proposes a network that predicts the next reading and compares it with the ground truth to detect anomalies based on a dynamic threshold. Despite the promising results, the model was only tested with the SMAP (Hundman *et al.*, 2018) dataset, so its generalization capabilities to other datasets remain unexplored.

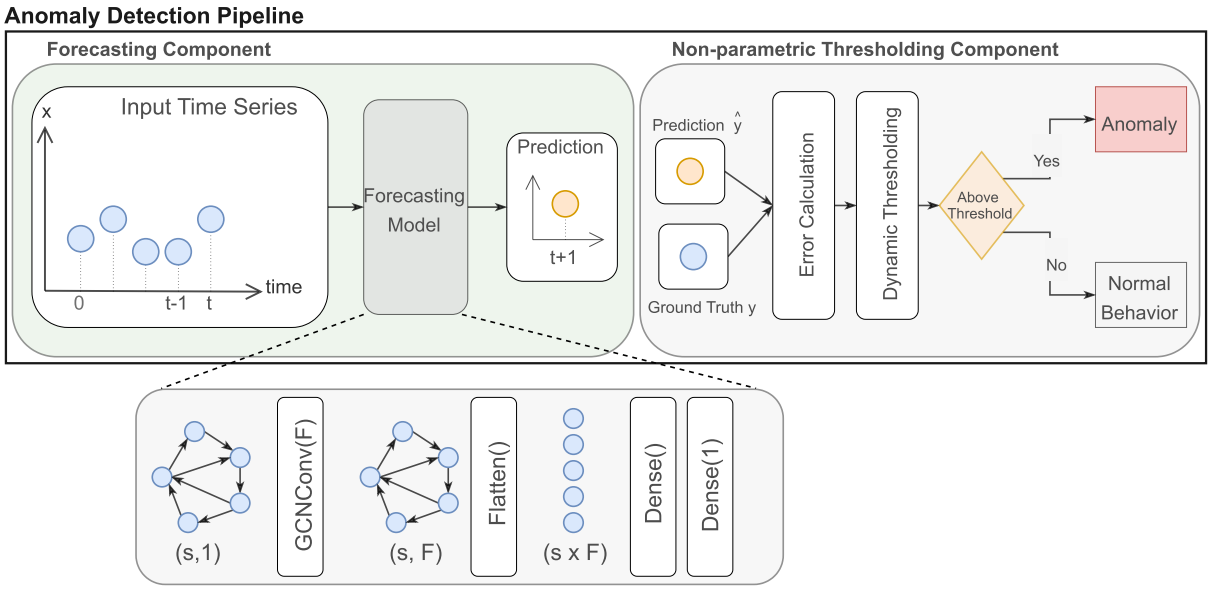

Figure 2: Process pipeline proposed by (Kiprit *et al.*, 2024).

The presented excerpt demonstrates the potential of machine learning models, particularly neural networks, in predicting satellite failures. However, there are still several challenges that are yet to be fully addressed:

- Model Selection: Various works have explored different architectures, such as LSTM, Real NVP, and GNNs; supporting the idea that there is no one-size-fits-all solution or yet an optimal architecture for this specific application.
- Generalization: Developing models that can perform well across different satellite missions remains a significant challenge due to the varying characteristics of each mission. Many existing works explore this aspect by testing their models on multiple datasets and comparing their performance. However, it is noted that selecting appropriate thresholds for anomaly detection is not only crucial but also highly application-specific, requiring careful tuning that may vary between different satellite missions. In that sense, further research is needed to develop models that can generalize effectively across various scenarios, especially for new missions not seen during training.
- Hardware Constraints: Deploying well-performing models on edge devices with limited computational resources, memory, and power consumption is a critical aspect that needs to be considered. To address this challenge, model optimization techniques such as quantization, pruning, and knowledge distillation can be employed to reduce model size and inference time without significantly compromising performance. Moreover, hardware accelerators and FPGA implementations are promising avenues to explore in this regard, since they can provide the necessary computational power while adhering to the constraints of satellite On-Board Computers (OBCs).
- Data Quality: Despite the availability of the ESA-ADB dataset, the quality of the data remains a concern. This is not limited to this dataset but is a common issue in real-world satellite telemetry data. The data may include noise, missing values, various sampling rates, and imbalanced classes due to the rarity of failure events. These factors can significantly compromise the development process of robust machine learning models.

Therefore, effective data preprocessing techniques, such as data cleaning, normalization, and feature selection, and data augmentation methods to inject synthetic anomalies, should be employed to enhance the quality of the dataset and improve model performance.

3. Methodology

The primary objective is to develop a neural network model specifically designed for time series anomaly detection in satellite telemetry data. The model will be initially trained and validated using the ESA-ADB dataset as the primary benchmark, with plans to extend testing to other publicly available datasets to evaluate cross-dataset generalization capabilities. A critical aspect of this project is that the final model must be optimized for deployment on embedded devices compatible with satellite On-Board Computers, taking into account the stringent constraints of space-grade hardware including limited computational resources, power consumption, and radiation tolerance.

The methodology will involve several key steps:

1. Data Exploration: Analyze the dataset to understand its structure, identify relevant features, and assess data quality.
2. Data Preprocessing: Clean the data by handling missing values, normalizing features, and choose the statistically relevant features for model training.
3. Model Design and Training: Design a neural network architecture suitable for time series data, such as LSTM or Transformer-based models. Train the model using appropriate loss functions and optimization techniques.
4. Model Evaluation: Evaluate the model's performance using metrics such as accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC).
5. Fine-tuning and Hyperparameter Optimization: Perform grid-search or other hyperparameter optimization methods to fine-tune the model for better performance.
6. Failure Injection Testing: Simulate various failure scenarios to assess the model's robustness and ability to detect anomalies.
7. Hardware/Software Deployment: Optimize the model for deployment on edge devices, considering constraints such as memory, computation power, and energy consumption.
8. Generalization Testing: Test the model on different satellite missions to evaluate its generalization capabilities.

4. Schedule

The Master's program requires the completion of at least 14 credits through coursework. The Table 1 lists the courses planned to be taken during the program.

Code	Name	Credits
MIC01	Física de Dispositivos Semicondutores	4
MIC02	Concepção de Circuitos em Alta Escala (VLSI)	4
MIC06	Seminários de Micro e Nanoeletrônica	1
MIC07	Seminários de Micro e Nanoeletrônica II	1
MIC09	Introdução a Aceleradores de Hardware para High Performance Computing	2
MIC91	Trabalho Individual I	1
MIC92	Trabalho Individual II	1
MIC98	Dissertação de Mestrado	

Table 1: List of courses planned for the Master's program, including the mandatory and suggested ones.

It is worth noting that MIC01 was already completed during the undergraduate engineering physics program, reducing the overall credit load (assuming it will be recognized). The coursework is planned to be concentrated in the first two semesters, allowing the final year to be dedicated primarily to research activities and dissertation writing.

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